

Md Mustakin Alam

Ph.D. Student in Computer Science
University of Louisiana at Lafayette
School of Computing and Informatics

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RESEARCH SUMMARY

My research focuses on designing efficient and adaptive machine learning and deep learning systems. My work spans efficient neural network training and intelligent AI architectures, with applications in federated learning, natural language processing, and computer vision. Through my research, I aim to improve the efficiency and practical deployment of modern AI systems.

EDUCATION

Ph.D. in Computer Science

Aug 2024 – May 2028

University of Louisiana at Lafayette

GPA: 4.00 / 4.00

Master of Science in Computer Science

Aug 2024 – May 2026

University of Louisiana at Lafayette

GPA: 4.00 / 4.00

Bachelor of Science in Computer Science and Engineering

May 2019 — May 2023

BRAC University

CGPA: 3.96 / 4.00

PUBLICATIONS

*Author/co-author of 11 peer-reviewed publications with 76 citations, h-index 5, and i10-index 3 (Google Scholar), including publications in top-tier venues such as **ICLR** and **ACM Multimedia**.*

Selected Conference Publications

1. **Md Mustakin Alam**, Shaker Islam, Aminul Islam. “**Batch Pruning by Activation Stability**”. In: The Fourteenth International Conference on Learning Representations, **ICLR 2026** - Top-Tier (A*) conference.
2. Aminul Islam, **Md Mustakin Alam**, Shaker Islam. “**The Birth of Vision Language**”. In: 33rd ACM International Conference on Multimedia, **ACMMM 2025** - Top-Tier (A*) conference.

Journal Publications

1. **Md Mustakin Alam**, Tanjim Ahmed, Meraz Hossain, Mehedi Hasan Emo, Md. Kausar Islam Bidhan, Md Tanzim Reza, Md. Golam Rabiul Alam, Mohammad Mehedi Hassan, Francesco Pupo, Giancarlo Fortino. “**Federated Ensemble-Learning for Transport Mode Detection in Vehicular Edge Network**”. In: Future Generation Computer Systems, **FGCS (2023)** - Q1 Journal.

Other Peer-Reviewed Publications

1. SweetCoat-2D: Two-Dimensional Bangla Spelling Correction and Suggestion Using Levenshtein Edit Distance and String Matching Algorithm. IJCNN 2023.
2. Early Autism Disorder Detection Through Visualizing Eye-Tracking Patterns Using Compact Convolutional Transformers. ICCTA 2023.
3. Ensemble of Gated Recurrent Unit and Convolutional Neural Network for Sarcasm Detection in Bangla. CCWC 2023.
4. Transformation of Visual Information into Bangla Textual Representation. CCWC 2023.
5. Optimization of Deep CNN-based Bangla Sign Language Recognition using XGBoost classifier. ICCTA 2023.
6. Consumer Review Analysis using NLP and Data Mining. CCWC 2023.
7. Bidirectional LSTM and NLP Based Sentiment Analysis of Tweets. SoCPaR 2022.
8. Human Speech Emotion Recognition Using CNN. ICCIT 2022.

Manuscripts Under Review

1. **Md Mustakin Alam**, Shaker Islam, Aminul Islam. “**Federated Data Pruning via Activation Stability**”. *Under Review*. 2026.
2. Gabriel Christian Trahan, **Md Mustakin Alam**, Aminul Islam. “**Disorder Over Order: Why Curriculum Learning Fails in Image Classification**”. *Under Review*. 2026.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

June 2026 – Present

University of Louisiana at Lafayette, USA

- Advised by Dr. Aminul Islam
- Conducting research on adaptive and efficient AI architectures

Machine Learning Engineer

June 2023 – July 2024

mPower Social Enterprises Ltd., Bangladesh

- Developed machine learning models for real-world applications
- Built LLM-based chatbot and question generation systems
- Developed pregnancy risk prediction model using healthcare data
- Worked on speech-to-text systems by fine-tuning Whisper models

TEACHING EXPERIENCE

Graduate Teaching Assistant

August 2024 – May 2026

University of Louisiana at Lafayette, USA

- Course: CMPS340 Design and Analysis of Algorithms by Dr. Aminul Islam
- Provided consultations during office hours
- Graded homeworks and exams

Adjunct Lecturer
BRAC University, Bangladesh

September 2023 – July 2024

- Taught Courses:
 - CSE422 Artificial Intelligence (Theory)
 - CSE422 Artificial Intelligence (Laboratory)
 - CSE341 Microprocessors (Laboratory)
 - CSE461 Introduction to Robotics (Laboratory)
- Supervised student research and projects

Undergraduate Teaching Assistant
BRAC University, Bangladesh

February 2022 – April 2023

- Assigned Courses:
 - CSE110 Programming Language I
 - CSE111 Programming Language II
 - CSE424 Pattern Recognition
 - CSE431 Natural Language Processing
 - CSE449 Parallel, Distributed, and High-Performance Computing
- Graded homeworks
- Provided consultation hours for students and assisted them with problem-solving and exam preparation
- Introduced students to research methodology and guided them in writing research papers

HONORS AND AWARDS

- Graduate Student Supplements (GRASS) Award – NSF & Louisiana Board of Regents 2026
- Academic Excellence Award – University of Louisiana at Lafayette 2025 – 2026
- Graduate Student Research Showcase Poster Competition – 2nd place 2025
- Vice Chancellor's List – BRAC University (Multiple semesters) 2019 – 2023
- Dean's List – BRAC University 2021
- Bachelor's with Highest Distinction – BRAC University 2023

EXPERIENCE WITH RESEARCH GRANTS

Awarded

- Graduate Student Supplements (GRASS) Award,
NSF EPSCoR & Louisiana Board of Regents 2026

Grant Proposals

- Vision Language, NSF Future CoRe. *Under Review* September 2025
- Energy-Efficient Deep Learning Training via Dynamic Pruning,
NSF Future CoRe. *Under Review* February 2026

ADVISING AND MENTORING

- Shaker Islam (Bachelor of Science, BRAC University; Incoming PhD student at Johns Hopkins University)
 - Co-authored articles at ACM MM 2025 and ICLR 2026.
- Sébastien Ricart and Lou-Anne Lecocq (Visiting Interns from CESI, France to UL Lafayette)

- Provided technical guidance and project mentorship under the supervision of Dr. Aminul Islam in 2025.
- Louis Covemacker and Hugo Kergoat (Visiting Interns from CESI, France to UL Lafayette)
 - Provided technical guidance and project mentorship under the supervision of Dr. Aminul Islam in 2024.
- A.S.M Mehedi Hasan, Md. Alvee Ehsan, Kefaya Benta Shahnoor, and Syeda Sumaiya Tasneem (Bachelor of Science, BRAC University)
 - Co-supervised undergraduate thesis in 2023.

ACADEMIC AND EXTRACURRICULAR SERVICES

- Conference Reviewer – HICSS 2026, NeurIPS 2026
- Journal Reviewer – Evolving Systems (Springer)
- Science Fair Judge – University of Louisiana at Lafayette (2025, 2026)
- Science Day Volunteer – University of Louisiana at Lafayette (2025)
- Student Mentor – BRAC University (2021 – 2022)
- Senior Executive – Robotics Club, BRAC University
- Assistant Director – Entrepreneurship Development Forum, BRAC University

PRESENTATIONS AND TALKS

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|------------------------|---|
| Conference Talk | Batch Pruning by Activation Stability, <i>The Fourteenth International Conference on Learning Representations (ICLR)</i> , Recorded Online, March 2026 |
| Poster | Energy-Efficient Deep Learning Training via Dynamic Data Pruning, <i>Graduate Student Research Showcase Poster Competition</i> , University of Louisiana at Lafayette, March 2026 |
| Poster | Vision Language, <i>Graduate Student Research Showcase Poster Competition</i> , University of Louisiana at Lafayette, March 2025 |
| Invited Talk | Vision Language, <i>Graduate Seminar</i> , University of Louisiana at Lafayette, March 2025 |
| Conference Talk | Bidirectional LSTM and NLP-Based Sentiment Analysis of Tweets, <i>14th Soft Computing and Pattern Recognition (SoCPaR'22)</i> , Online Presentation, December 2022 |